

IKEA E-Commerce Database Management System

CHAPTER 1

1. Introduction

The IKEA E-Commerce Database Management System is designed to manage the online operations of a furniture and home furnishing store. It enables customers to browse products, add items to their shopping cart, place orders, make secure payments, and track deliveries. The system also helps administrators manage products, inventory, customers, orders, warehouses, and shipping efficiently.

The project aims to demonstrate how a real-world e-commerce platform operates by integrating customer management, inventory management, order processing, payment handling, and logistics into a single database system.

1.1 Problem Statement

Managing an online furniture store involves handling thousands of products, customer records, inventory levels, orders, payments, and deliveries. Manual management of these operations can lead to errors, delays, duplicate records, inventory mismatches, and poor customer experience.

This project proposes a centralized database management system that automates these processes, improves data accuracy, reduces operational errors, and enhances customer satisfaction.

1.2 Stakeholders

The stakeholders involved in this project include:

- Customers
- Administrator
- Warehouse Manager
- Inventory Manager
- Delivery Partner
- Payment Gateway
- Customer Support Team
- Suppliers

1.3 Objectives

The primary objective of this project is to develop a reliable and efficient database management system for an online furniture shopping platform.

The system aims to:

- Manage customer information.
- Store product details.
- Process online orders.
- Manage inventory.
- Track deliveries.
- Generate reports.
- Improve overall business efficiency.

1.4. Key Objectives

- Develop a structured database for an online furniture store.
- Maintain accurate customer and product information.
- Automate order processing.
- Monitor stock availability.
- Ensure secure payment records.
- Improve order tracking and delivery management.
- Provide better customer experience.
- Reduce manual errors.

1.5. Business Requirements

The business requires a system that can:

- Allow customers to browse furniture products.
- Enable customer registration and login.
- Maintain detailed product catalogs.
- Manage shopping carts and wishlists.
- Process online orders.

- Handle secure payments.
- Track inventory across warehouses.
- Schedule product deliveries.
- Generate invoices.
- Collect customer reviews and ratings.
- Produce business reports for administrators.

1.6. Functional Requirements

The system shall:

- Register new users.
- Authenticate users through login.
- Display product categories.
- Allow product search and filtering.
- Add or remove products from the shopping cart.
- Place customer orders.
- Process online payments.
- Generate order confirmations.
- Update inventory after every purchase.
- Track shipment status.
- Allow customers to submit product reviews.
- Enable administrators to manage products, customers, and orders.

1.7. Non-Functional Requirements

The system should:

- Provide high security for customer data.
- Deliver fast response times.
- Ensure high availability.
- Support scalability for future expansion.
- Maintain data consistency.

- Provide an easy-to-use interface.
- Ensure database reliability.
- Support regular backup and recovery.

1.8. Motivation

The motive of this project is to understand the working of a real-world e-commerce platform by designing and implementing a database management system. It helps in learning database design, data relationships, SQL operations, inventory management, order processing, and business workflows. The project also develops practical knowledge of software development and prepares students for industry-level applications.

1.9. Conclusion

The proposed IKEA E-Commerce Database Management System provides a structured, automated framework designed to replace inefficient manual retail operations with a centralized, data-driven ecosystem. By successfully aligning the goals of diverse stakeholders ranging from customers and administrators to inventory managers and third-party logistics partners the database establishes a secure and scalable backend capable of supporting real-time transactions. Ultimately, this system ensures data integrity, minimizes operational errors, and optimizes tracking from warehouse to doorstep, laying down a reliable foundation for a modern, high-performing digital furniture marketplace.